M Stars

Fundamental Mode

Assume logMdot = $a_0 + a_1$ logM + a_2 logL + a_3 logR + a_4 logP. Use functions logP(logM,logR) and logR(logL, logM) to change this to the form logMdot = $a + b*$logL – c logM

First, M stars in the fundamental mode.
logR = -0.241(9) + 0.690(3)logL - 0.251(4)logM (R^2 = 0.990)
logP = -2.234(5) - 0.780(2)logM + 2.027(2)logR (R^2 = 0.999)
logMdot= -3.478 + 1.822 logM - 2.439logL - 1.302logR + 3.706logP (R^2 = 0.206)

```
In[6]:= Expand[Simplify[-3.478 + 1.822 logM - 2.439 logL - 1.302 (-0.241 + 0.690 logL - 0.251 logM) +
    3.706 (-2.234 - 0.780 logM + 2.027 (-0.241 + 0.690 logL - 0.251 logM))]]
Out[6]= -13.2538 + 1.84594 logL - 2.62741 logM
```

This suggests logMdot = -13.25 + 1.845 logL - 2.627 logM...

Overtone Mode

Assume logMdot = $a_0 + a_1$ logM + a_2 logL + a_3 logR + a_4 logP. Use functions logP(logM,logR) and logR(logL, logM) to change this to the form logMdot = $a + b*$logL – c logM

First, M stars in the overtone mode.
logR = -0.241(9) + 0.690(3)logL - 0.251(4)logM (R^2 = 0.990)
logP = -1.55428(1) - 0.528686(6)logM + 1.570244(6)logR (R^2 = 1)*
logMdot= -5.039 + 1.938 logM - 2.622logL - 1.081logR + 5.107logP (R^2 = 0.343)
*Test Data to confirm return of results from others' work

```
In[7]:= Expand[Simplify[-5.039 + 1.938 logM - 2.622 logL - 1.081 (-0.241 + 0.690 logL - 0.251 logM) +
    5.107 (-1.55428 - 0.528686 logM + 1.570244 (-0.241 + 0.690 logL - 0.251 logM))]]
Out[7]= -14.6488 + 2.16538 logL - 2.5035 logM
```

This suggests logMdot = -14.64 + 2.165 logL - 2.503 logM...

C Stars

Fundamental Mode

Assume logMdot = $a_0 + a_1$ logM + a_2 logL + a_3 logR + a_4 logP. Use functions logP(logM,logR) and logR(logL, logM) to change this to the form logMdot = $a + b*$logL – c logM

First, M stars in the fundamental mode.
logR = -0.323(27) + 0.737(7)logL - 0.371(14)logM (R^2 = 0.907)

logP = -2.285(7) - 0.826(4)logM + 2.043(3)logR (R^2 = 0.998)

logMdot= -5.141 + 2.859 logM - 3.201logL - 1.759logR + 5.791logP (R^2 = 0.421)

In[8]:=
```
Expand[Simplify[-5.141 + 2.859 logM - 3.201 logL - 1.759 (-0.323 + 0.737 logL - 0.371 logM) +
    5.791 (-2.285 - 0.826 logM + 2.043 (-0.323 + 0.737 logL - 0.371 logM))]]
```

Out[8]=
```
-21.6267 + 4.22207 logL - 5.66108 logM
```

This suggests logMdot = -21.62 + 4.222 logL - 5.661 logM...

Overtone Mode

Assume logMdot = $a_0 + a_1$ logM + a_2 logL + a_3 logR + a_4 logP. Use functions logP(logM,logR) and logR(logL, logM) to change this to the form logMdot = $a + b*$logL – c logM

First, M stars in the overtone mode.

logR = -0.323(27) + 0.737(7)logL - 0.371(14)logM (R^2 = 0.907)

logP = -1.55431(1) - 0.528676(6)logM + 1.570255(6)logR (R^2 = 1)*

logMdot= -5.204 + 2.626 logM - 2.547logL - 1.170logR + 4.857logP (R^2 = 0.382)

*Test Data to confirm return of results from others' work

In[9]:=
```
Expand[Simplify[-5.204 + 2.626 logM - 2.547 logL - 1.170 (-0.323 + 0.737 logL - 0.371 logM) +
    4.857 (-1.55431 - 0.528676 logM + 1.570255 (-0.323 + 0.737 logL - 0.371 logM))]]
```

Out[9]=
```
-14.8388 + 2.21161 logL - 2.33723 logM
```

This suggests logMdot = -14.83 + 2.211 logL - 2.337 logM...

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